2 **(a)** By reference to Newton's laws of motion, explain why when two particles, which are isolated, collide, the total momentum of the particles is conserved.

.....

.....

.....

.....

.....[3]

(b) A ferryboat of mass M_1 = 7.0 x 10^5 kg moves toward a docking bumper of mass M_2 that is attached to a shock absorber, as shown in Fig. 2.1.

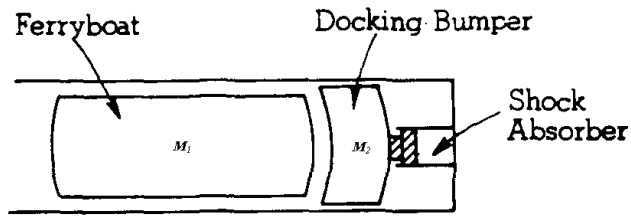

Fig. 2.1

Just before hitting the bumper, the ferryboat is travelling at a speed v = 3.0 m s^{-1}. After colliding inelastically with the bumper, the ferryboat and bumper move together with an initial speed of 2.0 m s^{-1}. The ferryboat comes to a complete stop 4.0 s after the collision. Fig. 2.2 shows the variation with time t of the velocity v of the ferryboat, where t = 0 s is the instant of collision.

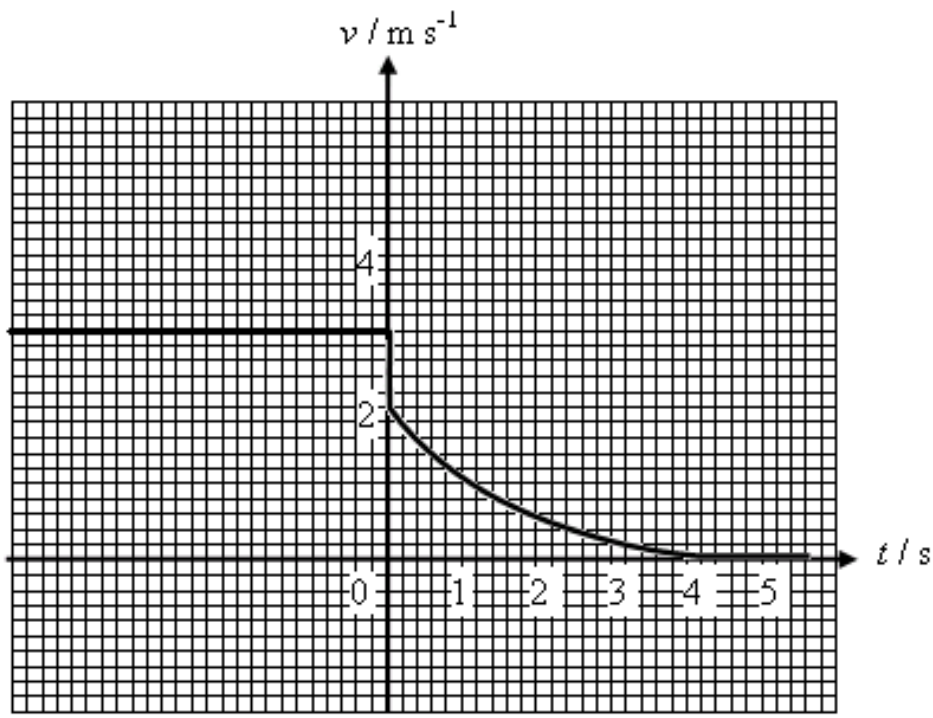

Fig. 2.2

[Turn over

(b) **(i)** Explain what is meant by *an inelastic collision*.

…….

.....[1]

(ii) Calculate the mass of the bumper M_2. State an important assumption that you have made in the calculation.

mass = …… kg [2]

(iii) Use the graph to determine the acceleration of the ferryboat at t = 1.0 s. Show your working on the graph clearly. State the direction of the acceleration.

acceleration = ……. m s^{-2}

direction = ……. [3]

(iv) Hence calculate the retarding force on the ferryboat at t = 1.0 s.

force = ……. N [1]

[Turn over